MILITARY
SPECIFICATION

MIL-C-14470A(SigC)
27 January 1961

Superseding
MIL-C-14470(SigC)
11 February 1957

CONVERTER, SIGNAL DATA CV-395( )/U

1. SCOPE

1.1 This specification covers one type of rack-mounted equipment designated as Converter, Signal Data CV-395( )/U, which includes Cable Assembly, Power, Electrical CX-4538( )/U. This equipment provides visual means for accurately tuning frequency Shift Converter CV-116/URR and radio receivers associated with it. (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder; Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.
PPP-B-636	Boxes, Fiber.
PPP-T-76	Tape; Pressure-Sensitive, Adhesive, Paper, Water-Resistant.
PPP-T-97	Tape; Pressure-Sensitive, Adhesive, Filament Reinforced.

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MIL-E-75	Packaging, Packing and Container Marking of Electron Tube.
MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior Packaging.
MIL-V-173	Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment.

FSC 5815

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SPECIFICATIONS (Contd)

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MIL-C-3432	Cable; Flexible and Extra Flexible, 300 and 600 Volts.
MIL-C-3767	Connectors, Electrical (Power Blade Type), General Specifications for.
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical (For use in Electronic, Communication and Associated Electrical Equipment).
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

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MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage (Pg 14 and 47).
MIL-STD-169	Extreme Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-B-61578	Grounding to Chassis.
SC-DL-166895	Converter, Signal Data CV-395( )/U.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description.- Converter, Signal Data CV-395( )/U is a tuning indicator which provides a means of accurately tuning Frequency Shift Converter CV-116/URR and associated radio receivers for reception of time division multiplex frequency shift radioteletype (rtt) signals.

3.2 Construction.- Converter, Signal Data CV-395( )/U shall be constructed in accordance with Drawing SC-DL-166895.

3.3 Plastic materials and parts.- Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.4 Wiring and cabling.- Wiring and cabling shall be neat and sturdy.

3.4.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnection of other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected.

(d) To ensure freedom of motion of lugs or terminals normally intended to have some degree of motion. (For example: Floating contacts on electron tube sockets).

(e) To facilitate field repair of broken or cut wires.

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3.4.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces, or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal partitions, shields or similar items less than 1/8 inch in thickness, the wires shall be protected by suitable grommets or bushings.

3.4.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.4.4 Connections.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Electrical connections shall not be made by clamping between a metallic and a nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.4.5 Grounding.- Ground connection to shields and other mechanical parts, except the chassis or frame, shall not be made to complete electrical circuits but only to eliminate high-potential ac points. Grounding to the chassis shall conform to Drawing SC-B-61578.

3.5 Soldering.-

3.5.1 Solder.- Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition Sn60, type AR or S, shall be used for general purposes. When type S solder is used for soldering electrical connections, a rosin type flux shall be employed and shall meet the applicable requirements of Specification QQ-S-571 for the core of type AR solder.

3.5.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.5.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign

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3.6 Cleaning.-

3.6.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on the chassis.

3.6.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.7 Marking.-

3.7.1 General.- Marking shall conform to Specification MIL-M-13231. (See 4.4.) Front panel marking shall be Group I as described in that specification.

3.7.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial numbers.- Converter, Signal Data CV-395( )/U shall have serial numbers.

3.8 Tropicalization of material.- Material shall be tested as follows:

3.8.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.8.2 Toxicity.- Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the unit or when carrying, operating, or maintaining the unit, or in use of the finished items when used for the purpose intended.

3.8.3 Flexibility.- Treatment shall not affect flexibility of treated materials, to the extent that the unit may fail to meet specified requirements when subjected to specified service conditions.

3.8.4 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2)

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3.9 Finish, protective.- The equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be final film E, semigloss, light-gray enamel, conforming to Specification MIL-F-14072.

3.10 Tubes, electron.- The equipment shall be furnished with the same tubes used when the set was inspected. The tubes shall remain in their respective sockets occupied during inspection of the units in which they are used, unless separate packaging is specified.

3.11 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the radio receiver. (See 4.16). Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.12 Cable Assembly, Power, Electrical CX-4538( )/U.- One power cord shall be supplied with each equipment. This cord shall be Cable Assembly, Power, Electrical CX-4538( )/U (6 ft +3 in -0 in (including terminations)) constructed as follows: A length of cord, electrical, Signal Corps Type CO-130 per Specification MIL-C-3432 shall terminate in a connector plug, two contact, male, flat, straight type UP 120M, per Specification MIL-C-3767, and Connector Plug, Amphenol AN/3106A-14S-7S or equal with Amphenol Cable Clamp AN/3057-6, or equal on the other end. One conductor of the cable shall be connected to terminal "A" and the other to terminal "C" of the three terminal Connector Plug AN/3106A-14S-7S. No connection shall be made to terminal "B" of this connector. The cable assembly shall meet the requirements and tests of Specification MIL-C-3885. (See 4.19).

3.13 Electrical requirements.-

3.13.1 Definitions.- The definitions of certain words and phrases as used in the specification are as follows:

Mark (M):	Any positive voltage input appearing at the signal input terminal.
Space (S):	Any negative voltage input appearing at the signal input terminals.
Reversals:	Alternate positive and negative square wave voltage pulses each of which is equal in duration. Unless otherwise specified, the amplitude of the voltages shall be considered symmetrical about zero.

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3.13.1 Definitions.- (Contd)

Mark/Space (m/s) Ratio:	The ratio of time duration of a positive voltage pulse to the time duration of a negative voltage pulse.
Space/Mark (S/M) Ratio:	The ratio of time duration of a negative voltage pulse to the time duration of a positive voltage pulse.

3.13.2 Operational performance.- The equipment shall be capable of operating with Frequency Shift Converter CV-116/URR. The tuning indicator shall receive information in the form of positive and negative square wave voltage pulses of varying lengths from the automatic frequency control discriminators of Frequency Shift Converter CV-116/URR, and shall convert this information into different voltage values which are substantially independent of pulse lengths of the input signal. These new voltage values shall be used to operate a zero center meter which shall indicate zero (correct tuning) when the amplitude of the positive voltage input is equal to the amplitude of the negative voltage input. Inequalities in the peak values of the two input voltages shall cause the tuning indicator meter to assume a steady deflection to the right or left of zero. A three position switch in the input circuit of the tuning indicator shall permit monitoring of either channel A, or channel B of Frequency Shift Converter CV-116/URR, or an internal calibrating a-c voltage. (See 4.15.5).

3.13.3 Power source.- The equipment shall operate from a power source of 115 volts alternating current, single phase 50-60 cycles, and shall perform in compliance with the requirements of this specification. Supply voltage and frequency variations shall not exceed ± 10% and ± 5%, respectively.

3.13.4 Input impedance.- The input impedance of both input channels shall be not less than one megohm.

3.13.5 Tuning indicator.

3.13.6 Calibrate adjustment.- With the channel selector switch in position "C", it shall be possible to set the tuning meter to zero by adjustment of the calibration control, thus compensating for tolerances in electron tube characteristics and other component tolerances in the equipment. A clockwise rotation of the calibration control shall result in a tuning meter deflection to the right; a counterclockwise rotation shall result in a tuning meter deflection to the left.

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3.13.7 Tuning indication with reversals.- When reversals having a peak-to-peak amplitude of 24 volts (M +12V; S-12V) are applied to the input terminals, the tuning meter shall indicate zero (±1) division on a 100-0-100 scale with the channel selector switch in either channel "A" or channel "B" position.

3.13.8 Tuning indications with reversals, mark and space voltages symmetrical.- Tuning meter indication shall be within the limits indicated below when operated under conditions specified in the following table: (See 4.10.1).

SIGNAL INPUT VOLTAGE		TUNING METER INDICATION
MARK	SPACE	DIVISIONS OF 0-100 SCALE
24	0	+40 ±10
18	6	+20 ±5
12	12	0 ±1
6	18	-20 ±5
0	24	-40 ±10

3.13.9 Tuning indications with signals having M/S and S/M Ratios greater than unity.- The tuning meter indication shall be within the limits indicated below when operated under conditions specified in the following table: (See 4.10.2).

SIGNAL INPUT VOLTAGE		TUNING METER INDICATION	
MARK	SPACE	M/S = 9/1	S/M = 9/1
24	0	+50 ± 10	+30 ± 10
18	6	+30 ± 5	+10 ± 5
12	12	+10 Max	-10 Max
6	18	-10 ± 5	+30 ± 5
0	24	-30 ± 10	-50 ± 10

3.13.10 Steady state mark or space tuning meter indication.- A steady mark signal of plus 12 volts dc shall cause the meter to indicate 45 ± 10 to the right of zero. A steady space signal of minus 12 volts dc shall cause the meter to indicate 45 ± 10 to the left of zero. After switching from one steady state to another, the tuning meter shall come to rest not sooner than 10 seconds after the switching operation. (See 4.10.3).

3.14 Service conditions.- The equipment shall meet the following service conditions:

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3.14.1 Operation.- Continuous use for a period of one year under the worldwide environmental conditions specified in 3.14.2 through 3.14.6 with a duty cycle of 24 hours on and 0 (zero) hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample (see 3.17) shall be considered as compliance with this requirement.

3.14.2 Temperature.- (See 4.14).

(a) Operating: Ambient temperature in the range of + 150°F to -40°F. (The 150°F temperature includes effect of sun-load). Exposure at high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80° F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours at any one time.

3.14.3 Relative humidity (see 4.15).- Up to 97 percent relative humidity, with condensation, for 20 hours; and exposure at 100 percent relative humidity, with condensation for 4 hours.

3.14.4 Elevation (see 4.17).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 25,000 feet above sec level.

3.14.5 Orientation.- (See 4.18)

(a) Operating: Any orientation up to 20 degrees from normal operating position (that is: forward, backward, left or right).

(b) Nonoperating: Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of preproduction sample (see 3.17) shall be considered as compliance with this requirement.

3.14.6 Vibration, bounce and shock.- As described in 3.15.

3.15 Vibration, bounce and shock.-

3.15.1 Vibration, internal.- The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.11).

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3.15.2 Bounce.- The equipment shall meet full specification performance requirements, and there shall be no physical damage except surface abrasion after subjection to the test of 4.12.

3.15.3 Shock, bench-handling.- The equipment shall meet full specification performance requirements, and there shall be no physical damage after subjection to the test of 4.13.

3.16 Preconditioning.- The equipment shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.12).

3.17 Preproduction samples.- The contractor shall furnish preproduction samples for approval as required by the invitation for bids and contract. (See 6.2).

3.18 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spares shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2).

3.19 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.3 Plastic materials and parts.
3.4 Wiring and cabling.
3.4.1 Slack
3.4.2 Protection
3.4.3 Splicing
3.4.5 Grounding
3.5 Soldering
3.5.2 Acid and acid salts
3.5.3 Process
3.6 Cleaning
3.8 Tropicalization.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3).

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5).

(2) Acceptance inspection of preparation for delivery. (See 4.20).

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I and the inspection specified for groups A and B. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling, and (4) moisture resistance; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I - Preproduction inspection

Inspection	Rqts par.	Insp par.
Temperature	3.14.2	4.14
Moisture resistance	3.14.3	4.15
Vibration	3.15.1	4.11
Bounce	3.15.2	4.12
Shock, benchhandling	3.15.3	4.13
Elevation	3.14.4	4.17
Orientation	3.14.5	4.18

4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Marking	3.7
Finish	3.9

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4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion, may perfo.m all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6.) Each unit which will be subjected to group A and group B inspection shall be preconditioned after final assembly. (See 3.16.)

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.15.5) shall be last. (See 6.4.)

Table II - Group A Inspection

Examination or Test	Rqt par.	Insp par.	AQL Major	AQL Minor
Converter				
Visual and mechanical	3.19	4.7	1.5 dphu*	6.5 dphu
Electrical				
Calibrate adjustment	3.13.6			
Tuning indication with reversals	3.13.7, 3.13.8	4.10.1	1.5 dphu for the group	**
Tuning indicatio, M/S and S/M ratios	3.13.9	4.10.2		
Steady state	3.13.10	4.10.3		
Operational	3.13.2	4.15.5	1%	
Cable Assembly	3.12	4.19	As per MIL-E-3885	

* dphu - Defects per hundred units.
** All electrical defects are considered major category.

4.5.2 Group B inspection.- This inspection, including sampling, shall be performed on the interchangeability requirement, 3.11, in accordance with 4.16. Sampling shall conform to levels L-8 for normal and tightened, and L-6 for reduced inspection. The reduced inspection procedure shall be R-1. An AQL of 4% shall be used.

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4.6 Bounce preconditioning.- The unit shall be placed in its normal operation position on the table of the package tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.7 Visual and mechanical inspection.- The equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.8 Electrical tests.-

4.8.1 Standard test conditions.- Unless otherwise required by a specific test requirement, the tests specified herein shall be made under the following conditions:

(a) Power supply input shall be 115 volts ac, 60 cycles, single phase. (Note: Similar verification tests shall be conducted at specified high and low voltage and frequency ranges, to determine compliance with specified performance).

(b) Ambient temperature and humidity shall be that prevailing indoors, and in any case shall be at least +77 ±10F at normal existing relative humidity.

4.9 Test equipment.- Test equipment shall be as follows:

(a) A signal source capable of sending:

(1) Reversals at a rate of 75 ± 5 cycles per second with positive and negative voltages independently variable from 0 to 24 volts dc.

(2) Signals consisting of marks and spaces having a mark space ratio of 9:1 with positive and negative voltages variable from 0 to 24 volts dc. The length of the space pulse shall be 7 milliseconds (± 1 ms).

(3) Signals consisting of marks and spaces having Space/Mark ratio of 9:1 with positive and negative voltages variable from 0 to 24 volts dc. The length of the mark shall be 7 milliseconds (± 1 ms).

(b) Oscilloscope, Dumont #304 or equal for dc observations.

(c) Variac -0 -130 volts ac.

(d) Vacuum tube voltmeter, RCA Voltohmist WV-97A, or equal.

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4.10 Tests for tuning indication.-

4.10.1 Reversals.- The equipment shall be adjusted for zero meter indication with the channel selector switch in position "C". The channel selector switch shall then be placed in position "A", and signals corresponding to those specified in 3.13.8 shall be fed to the channel "A" input jack. Tuning meter indication shall be recorded for each of the 5 different signal voltage values specified. This test shall be repeated using ac input voltages to the equipment of plus 10 and minus 10 percent of nominal 115 volts ac. A single test at one signal input and one supply voltage shall be made with the channel selector switch in position "B" to ascertain that channel "B" is properly connected. (See 3.13.8).

4.10.2 Signals with M/S and S/M ratios greater than unity.- The equipment shall be adjusted for zero meter indication with the channel selector switch in position "C". The channel selector switch shall then be placed in position "A", and signals corresponding to those specified in 3.13.9 shall be fed to the channel "A" input jack. Tuning meter indication shall be recorded for each of the 5 different signal voltage values specified. This test shall be repeated using ac input voltages to the equipment of plus 10 and minus 10 percent of nominal 115 volts ac. (See 3.13.9.)

4.10.3 Steady state mark, or space tuning indication.- The equipment shall be adjusted for zero meter indication with the channel selector switch in position "C". The channel selector switch shall then be placed in position "A" and signals corresponding to those specified in 3.13.10 shall be fed to the channel "A" input jack. Tuning meter indication, and time required for the meter to come to rest after switching from one steady state condition to another shall be recorded. This test shall be repeated using ac input voltages to the equipment of plus 10 and minus 10 percent of nominal 115 volts ac. (See 3.13.10.)

4.11 Test for internal vibration.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.15.1 and 3.15.2:

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

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4.11 Test for internal vibration.- (Contd)

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.12 Bounce test.- The equipment shall be placed on the table of the package tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. During each 1/2 hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.13 Shock test; bench-handling.- The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing:

4.14 Temperature test for equipment.- The equipment shall be subjected to the temperature cycle of Standard MIL-STD-169. Measurements shall be taken as follows:

Paragraph	Step	Measurement	Acceptance Test Results
3.13.8, 3.13.10	1	Tuning indications	Full performance of 3.13.8 and 3.13.10
" "	2A	" "	" " "
" "	5	" "	" " "
" "	6A	" "	" " "
" "	8	" "	" " "
" "	3	" "	" " "
" "	10	" "	" " "

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4.15 Moisture and resistance test for equipment.

4.15.1 Test conditions.-

(a) Do not remove the equipment from the humidity chamber for measurement.

(b) Complete measurements as quickly as possible.

(c) Keep equipment secured within its case.

4.15.2 Test procedures.-

(a) Dry at +130 F ± 5 F for 24 hours.

(b) Condition at +77 ± 5 F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.10.1, 4.10.2 and 4.10.3 and readjust or realign as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48 hour cycles. Temperature, relative humidity and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurement specified in 4.10.1, 4.10.2 and 4.10.3 shall be made during the periods shown on the standard.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.15.3.

4.15.3 Performance.- The unit shall meet full specification requirements. There shall be no mechanical deterioration as evidenced by corrosion of metal parts, and there shall be no degradation in performance when tuning indication measurements of 3.13.8 and 3.13.10 are performed, from that obtained prior to cycling of 4.15.2(d).

4.15.4 Failure.- If any equipment fails to meet the performance specified in 4.15.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.15.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.15.2(e), it does not pass the test.

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4.15.5 Operational inspection.- (See 6.4) The equipment shall be tested to determine compliance with the requirements of 3.13.2.

4.16 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, and insertable assemblies when such dimensions affect mating of parts.

(b) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.17 Elevation test.- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment).

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
3.13.8	Tuning Indication
3.13.10	Tuning Indication

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.17(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment, and after stabilization, repeat the measurements specified in 4.17.(a). The equipment shall meet specified performance for these measurements.

4.18 Orientation test.- The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 +3 degrees. During inclination in each place, the equipment shall meet specified performance when tested in accordance with 4.10.

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4.19 Power Cord CX-4538( )/U.- The finished cord assembly shall meet the requirements of this specification and the applicable requirements and tests of Specification MIL-C-3885.

4.20 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.21 Rough handling test (preparation for delivery).- When rough handling test is required by the contract(see 6.2(e)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

(a) 3.11.3

(b) 3.11.5

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

5.1.1 Level A.- Each Converter CV-395( )/U shall be packaged in accordance with items 1 through 6 of Figure 1 and Table III.

5.1.2 Level C.- Each converter CV-395( )/U shall be packaged as in 5.1.1 except that items 3 through 6 shall be Type III, class 1, in accordance with Table III of the box specification. The box closure shall be made in accordance with the requirements in the appendix of the box specification.

5.1.3 Package performance.- Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling test and the cyclic exposure test will be required only when invoked by the bid request or contract (see 6.2).

5.2 Packaging.-

5.2.1 Level A.- Six each Converter CV-395( )/U shall be packed in accordance with items 7 through 9 of Figure 1 and Table III. Box closure shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.- Six each Converter CV-395( )/U shall b e packed in accordance with items 7 and 8 of Figure 1 and Table III, except the fiberboard box shall be Type 1, class 1 of Table III and the box closure shall be in accordance with the requirements in the appendix of the box specification.

5.2.3 Level C.- The pack shall be as specified in 5.2.2.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

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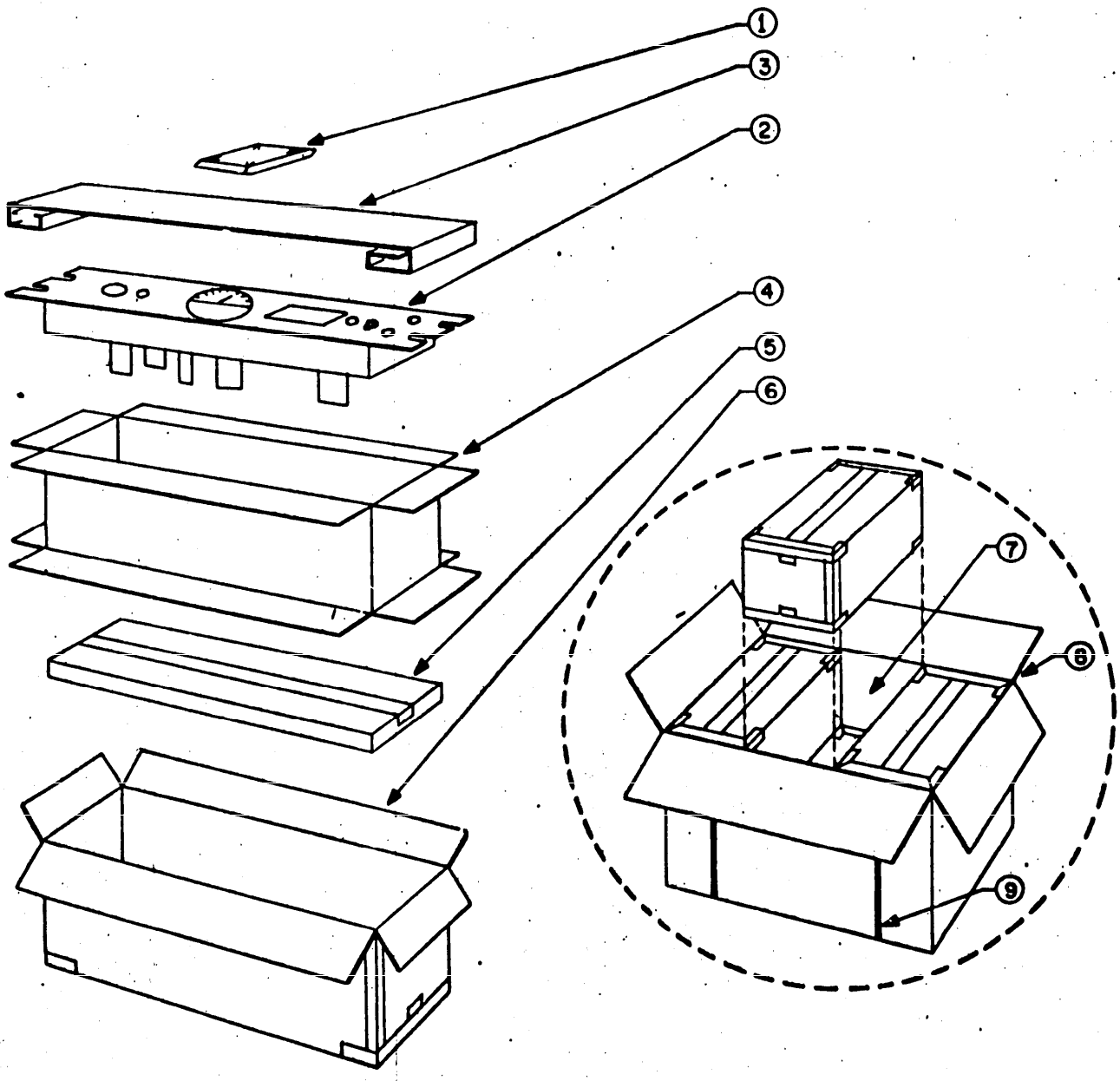
BILL OF MATERIAL
TABLE III

REF No.	ITEM Nomenclature	No Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENTS Spec No	T	G	C	S	Flute	PSI
1	TM., Bag	1	IC-6				MIL-B-117	II		d			
2	Converter	1	IC-5	19	6-1/4	3-1/2							
3	Cell; Top	1		20	6-3/4		PPP-B-636	I	6	2		C	
4	Sleeve	1		41-1/2	9		PPP-B-636	I	6	2		C	
5	One Piece Folder												
	Contains:	1		20	6-3/4	2	PPP-B-636	I	6	2		C	
	Lamps	1		2	1-1/2	1-1/4							
	Fuses	5		2	1-1/2	1-1/4							
	Tubes	2					MIL-E-75	Group		I			
	Tape Closure	1		26	2		PPP-T-76						
6	Box, Fiberboard	1		20	6-3/4	10	PPP-B-636	I	6	2		C	
	Tape Closure	2		26	2		PPP-T-76						
		4		11	2								
		1		10	2								
7	Package Item 6	6		3	2								
8	Fiberboard Box	1		31-3/4	14-1/4	20-1/2	PPP-B-636	I	3	2			
9	Strapping	2		70	1/2		PPP-T-97	III					

PACKED WEIGHT - 70
PACKED VOLUME - 5.7

LEGEND
T - Type S - Style
G - Grade C - Class

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NOTE: Fiberboard (Cells)(Pads)(Trays) will be Scored Across Corrugations

FIGURE 1

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6. NOTES

6.1 Intended use.- Converter, Signal Data CV-395( )/U is used for accurately tuning Frequency Shift Converter CV-116( )/URR, and associated receivers, for reception of time division multiplex frequency shift radioteletype signals.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type required.

(c) Levels of preservation, packaging and packing required.

(d) If rough handling test or cyclic exposure test is required, so state.

(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.17).

(f) Marking and shipping of samples

(g) Place of final inspection.

(h) Technical literature required. (See 3.18).

(i) Quantity of tools and running spare parts required. (See 3.18).

(j) Submission of the statement of treatment referenced in 3.8.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CV-395W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Location of operational inspection.- It is desirable that the operational inspection (4.15.5) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

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6.5.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.5.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.6. Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. Tho amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A-- The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B-- That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C-- A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.7 Dimensional data.- Sizes of packaging materials prescribed in Section 5 are based on the dimensions of the equipment cited on the applicable Bill of Material. When the dimensions of the equipment vary from those cited, the sizes of the packaging materials shall be adjusted accordingly. When shown in the Bill of Materials (see Section 5), corrugated fiberboard manufactured with A, B or C fluting may be used at the option of the contractor. When the fluting used is not the same as that cited in the Bill of Material, the dimensions of the affected packaging and packing materials will be adjusted accordingly.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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